

Practice Problem Set 2

ECON 101 – Macroeconomics

Fall 2025

1. The market for coffee is initially in equilibrium. Suppose research shows that drinking coffee reduces the risk of heart disease.
 - (a) Illustrate on a graph the impact on the demand curve.
 - (b) Show the new equilibrium price and quantity.
 - (c) Explain the changes.
2. In the market for smartphones, a new technology reduces production costs.
 - (a) Draw the original supply and demand curves.
 - (b) Show how the supply curve shifts.
 - (c) Indicate the effect on equilibrium price and quantity.
3. Suppose demand and supply in the market for tablets are given by:

$$Q_d = 100 - 2P \quad Q_s = 20 + 3P$$

where Q is quantity and P is price in dollars.

- (a) Find the equilibrium price and quantity.
 - (b) Determine whether there is a shortage or surplus if $P = 10$.
4. Consider the same market as in Problem 3. Suppose the government imposes a price floor of $P = 20$.
 - (a) Calculate the quantity demanded and supplied at this price.
 - (b) Determine the size of the surplus.
 - (c) Is this price floor binding? Why or why not?

5. In the housing rental market, demand and supply are given by:

$$Q_d = 80 - 2P \quad Q_s = 10 + 2P$$

If the government sets a rent ceiling at $P = 15$:

- (a) Find the equilibrium price and quantity without the ceiling.
- (b) Compute the shortage created by the ceiling.
- (c) Discuss one possible unintended consequence of this policy.

6. **Expenditure Approach**

Suppose in 2024, the following data for Canada are reported (in billions of dollars):

- Consumption = 1,400
- Investment = 500
- Government Purchases = 600
- Exports = 400
- Imports = 450

- (a) Calculate GDP using the expenditure approach.
- (b) If depreciation is \$200 billion and net factor payments from abroad are -\$50 billion, calculate **Net National Product (NNP)** and **Gross National Product (GNP)**.

7. **Income Approach**

A simplified economy reports the following for 2024 (in billions of dollars):

- Wages and Salaries = 2,200
- Rental Income = 300
- Corporate Profits = 400
- Net Interest = 200
- Indirect Taxes less Subsidies = 250
- Depreciation = 150

- (a) Compute GDP using the income approach.
- (b) Compare it to the expenditure method if GDP was \$3,350 billion. Are there any statistical discrepancies?

8. Real vs Nominal GDP

A two-good economy produces only bread and smartphones.

Year	Bread (Qty)	Price	Smartphones (Qty)	Price
2023	100	\$2	50	\$400
2024	110	\$2.20	60	\$380

- (a) Compute **Nominal GDP** for 2023 and 2024.
- (b) Using 2023 as the base year, compute **Real GDP** for 2023 and 2024.
- (c) Calculate the **GDP deflator** for 2024.

9. CPI Calculation

Assume a typical consumer basket contains:

- 10 loaves of bread
- 5 movie tickets

Prices are given below:

Year	Bread Price	Movie Ticket Price
2023	\$2	\$10
2024	\$2.20	\$12

- (a) Compute the **CPI** in 2023 (base year) and 2024.
- (b) Calculate the **inflation rate** between 2023 and 2024.
- (c) Compare the CPI inflation rate with the GDP deflator inflation rate from Question 3. Why might they differ?

10. Nominal vs Real Wages

A worker earns \$20 per hour in 2023. The CPI rises from 100 in 2023 to 110 in 2024.

- (a) If the worker's nominal wage in 2024 is \$21, compute the **real wage** in 2024 (base year = 2023).
- (b) Did the worker experience an increase in purchasing power?